How can we give an operational definition of the concept of distance? The answer to this question in the context of string theory is rather surprising. In 1988, the physicists Robert Brandenberger of Brown University and Cumrun Vafa of Harvard University pointed out that if the spatial shape of a dimension is circular, there are two different yet related operational definitions of distance in string theory. Each lays out a distinct experimental procedure for measuring distance and is based, roughly speaking, on the simple principle that if a probe travels at a fixed and known speed then we can measure a given distance by determining how long the probe takes to traverse it. The difference between the two procedures is the choice of probe used. The first definition uses strings that are not wound around a circular dimension, whereas the second definition uses strings that are wound. We see that the extended nature of the fundamental probe is responsible for there being two natural operational definitions of distance in string theory. In a point-particle theory, for which there is no notion of winding, there would be only one such definition.

How do the results of each procedure differ? The answer found by Brandenberger and Vafa is as surprising as it is subtle. The rough idea underlying the result can be understood by appealing to the uncertainty principle. Unwound strings can move around freely and probe the full circumference of the circle, a length proportional to R. By the uncertainty principle, their energies are proportional to 1/R (recall from Chapter 6 the inverse relation between the energy of a probe and the distances to which it is sensitive). On the other hand, we have seen that wound strings have minimum energy proportional to R; as probes of distances the uncertainty principle tells us that they are therefore sensitive to the reciprocal of this value, 1/R. The mathematical embodiment of this idea shows that if each is used to measure the radius of a circular dimension of space, unwound string probes will measure R while wound strings will measure 1/R, where, as before, we are measuring distances in multiples of the Planck length. The result of each experiment has an equal claim to being the radius of the circle—what we learn from string theory is that using different probes to measure distance can result in different answers. In fact, this property extends to all measurements of lengths and distances, not just to determining the size of a circular dimension. The results obtained by